

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General certificate of education ordinary level

COMBINED SCIENCE

4003/2

PAPER 2 Theory

JUNE 2019 SESSION

2 hours

Candidates answer on the question paper

Additional materials: Calculator (Optional)

Allow candidates 5 minutes to count pages before the examination.

This booklet should not be punched or stapled and pages should not be removed.

The Periodic Table is provided on page 20.

TIME 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.

Write your centre and candidate number in the boxes on the top right corner of every page of this paper.

Check if the booklet has all the pages and ask the invigilator for a replacement if there are duplicate or missing pages.

Write your answers in the spaces provided on the question paper.

Section A

Answer **all** questions.

Section B

Answer any **two** questions.

Section C

Answer any **two** questions.

Section D

Answer any **two** questions.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question.

This question paper consists of 20 printed pages.

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Section A

Answer all questions in this section in the spaces provided on the question paper.

1. a) State any **two** differences between the structure of an insect pollinated flower and a wind pollinated flower.

1.

2. [2]

- b) i) During photosynthesis, carbon dioxide and **X** react to produce glucose and oxygen.

Name the reactant **X**.

..... [1]

- ii) State any **one** condition needed for photosynthesis to take place.

..... [1]

- iii) Describe what happens to the glucose after its production.

.....

.....

.....

.....

..... [2]

2. a) State any **two** differences between sexual and asexual reproduction.

1.

2. [2]

- b) i) Distinguish between passive and active immunity.

.....

.....

..... [2]

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ii) Explain the term *natural immunity*.

.....
.....
.....

[2]

c) Name the organism which causes malaria.

.....

[1]

3. a) Methane is an alkane.

i) Name the two elements found in methane.

1
2

[2]

ii) State the type of bond formed between the elements named in (i).

.....

[1]

iii) Draw a dot and cross diagram to show the bonding in methane.

[2]

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- b) Explain why alkenes are more reactive than alkanes.

.....
..... [1]

4. Iron is produced in the blast furnace.

- a) Write the chemical formulae of any **two** raw materials which are fed into the blast furnace.

.....
..... [2]

- b) Stainless steel is an alloy of iron and two other metals.

- i) State the other **two** metals in stainless steel.

1
2 [2]

- ii) State **one** property of stainless steel and relate it to the use of stainless steel.

property
use
..... [2]

- c) State **one** way of protecting iron from rusting other than alloying it.

..... [1]

5. a) **Fig.5.1** shows a current carrying conductor.

Draw, on **Fig.5.1**, the magnetic field lines around the conductor.

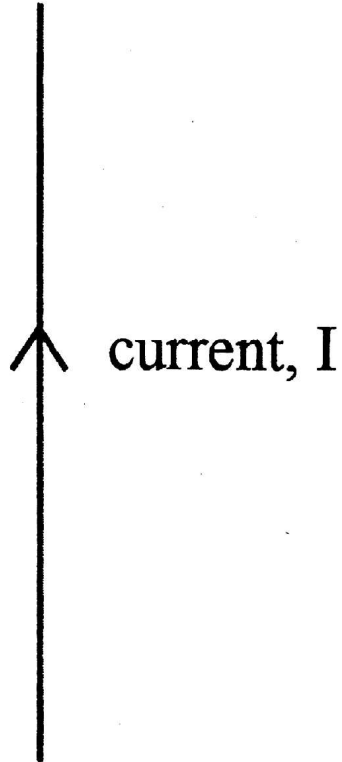


Fig.5.1

[2]

- b) The current carrying conductor is then placed between two magnets and connected to a circuit as shown in **Fig.5.2**.

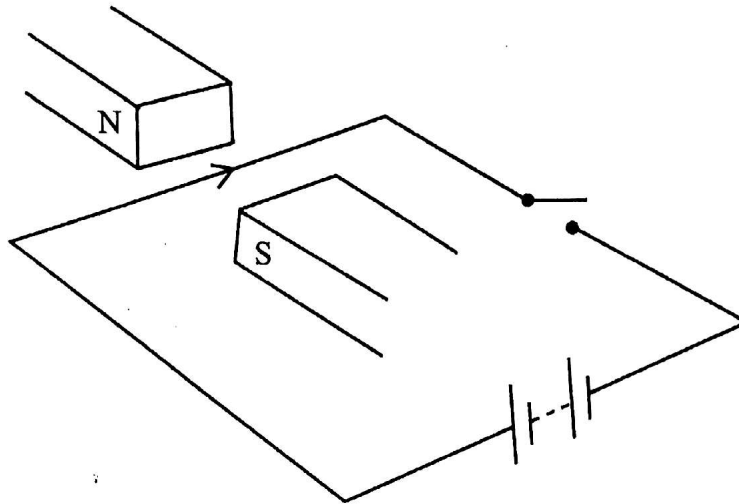


Fig.5.2

- i) State, with a reason, what happens when the

1. switch is closed,

.....
.....
.....

2. battery terminals are reversed and switch is closed.

.....
[4]

- ii) Give **one** application of the principle illustrated in **Fig.5.2**.

.....
[1]

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6. a) A box measuring 0.3 m wide, 0.5 m long and 0.6 m high has a weight of 20 N. The box rests on a table.

- i) Define the terms *weight* and *pressure*.

weight

pressure

[2]

- ii) Calculate the pressure exerted by the box when it rests on the 0.5 m and 0.6 m face.

[2]

- iii) Explain how the pressure calculated in (ii) compares with the pressure exerted when the same box rests on the 0.3 m by 0.5 m face.

.....

[2]

- b) Name the instrument used to measure fluid pressure.

.....

[1]

Section B

Answer any **two** questions. Write your answers in the spaces provided on the question paper.

7. a) i) State any **two** sexually transmitted infections.

1
2
[2]

- ii) Give the causative agent for each sexually transmitted infection named in (i).

.....
.....
[2]

- b) Describe and explain how cholera is treated.

.....
.....
.....
.....
[4]

- c) State any **two** effects of tobacco smoke on health.

.....
.....
[2]

8. a) i) State the **three** functions of blood.

1
2
3
[3]

- ii) Outline any **three** structural differences between arteries and veins.

.....
.....
.....
.....
.....
[3]

- b) Describe how plants are adapted to reduce water loss.

.....

.....

.....

.....

.....

[4]

9. a) Fig.9.1 shows the carbon cycle.

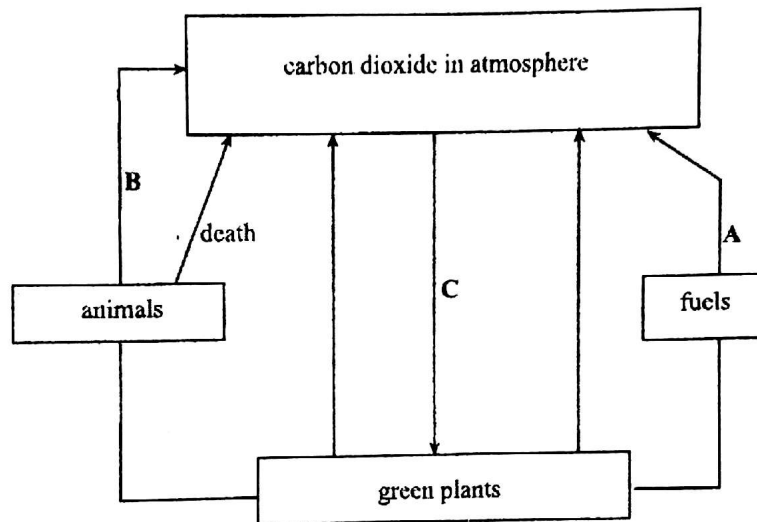


Fig.9.1

- i) Identify the processes labelled A and B.

A

B

[2]

- ii) Describe process C.

.....

.....

[2]

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- b) i) State **one** process which increases the amount of nitrogen in the atmosphere.

.....
[1]

- ii) State **two** processes which reduce the amount of nitrogen in the atmosphere.

.....
.....
[2]

- c) i) State any **two** problems caused by limited biodiversity.

.....
.....
[2]

- ii) Give any **one** advantage of biodiversity.

.....
.....
[1]

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Section C

Answer any **two** questions. Write your answers in the spaces provided on the question paper.

10. Fermentation of glucose solution produces dilute ethanol ($\text{C}_2\text{H}_5\text{OH}$).

- a) i) Name a physical process by which pure ethanol can be obtained from the dilute ethanol.

.....
[1]

- ii) Describe the process named in (i).

.....
[3]

- iii) State any **two** uses of ethanol.

1
2
.....
[2]

- iv) Calculate the molecular mass of ethanol.

[2]

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v) Calculate the percentage of carbon in ethanol.

[2]

11. a) The mass number of potassium is 39 and its proton number is 19.

i) State the number of electrons in the potassium atom.

..... [1]

ii) Determine the number of neutrons in the potassium atom.

.....
 [1]

b) Potassium reacts with fluorine by donating electrons.

i) State the number of electrons donated by potassium and the charge of the potassium ion.

number of electrons donated
 charge of the potassium ion [2]

ii) Write the formula of potassium fluoride.

..... [1]

iii) State any **one** physical property of potassium fluoride.

.....
 [1]

- c) Sodium hydroxide (NaOH) is dissolved in water to form a solution of concentration 0.5 mol/dm^3 .

Calculate the number of moles of sodium hydroxide in 250 cm^3 of the solution.

[4]

12. Table 12.1 shows the number of bubbles produced when metals A, B, C and D reacted with dilute sulphuric acid for three minutes.

Table 12.1

metal	number of bubbles after 3 minutes
A	6
B	3
C	0
D	14

- a) Name the gas produced.

[1]

- b) Identify, with a reason, the most reactive metal.

[2]

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c) The metals used were copper, calcium, lead and zinc.

i) State, with a reason, the letter which represents copper.

.....
..... [2]

ii) State any **one** alloy of copper.

..... [1]

iii) State the metal used for galvanising iron.

..... [1]

d) i) Define a compound.

.....
..... [1]

ii) Explain why copper does **not** react with zinc oxide.

.....
..... [2]

Section D

Answer any **two** questions. Write your answers in the spaces provided on the question paper.

13. a) **Fig.13.1** shows a pulley system used to lift a load of 600 N. The efficiency of the pulley system is 75%.

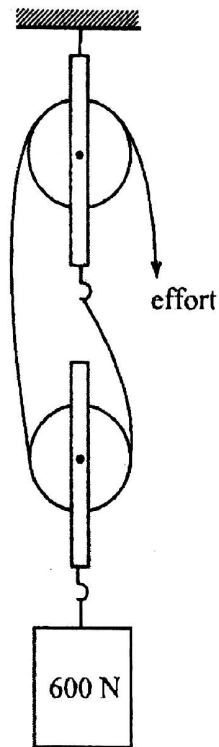


Fig.13.1

- i) Define the term *machine*.

.....
[1]

- ii) Calculate the mechanical advantage (MA) of the pulley system.

[3]

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- b) Electrical appliances which have a metal casing are usually fitted with an earth wire. Explain how an earth wire acts as a safety device.

.....

.....

..... [3]

- c) Define the term

- i) mass,

..... [1]

- ii) weight,

..... [1]

- iii) momentum.

..... [1]

14. a) Fig.14.1 shows how the length of a glass rod was measured.

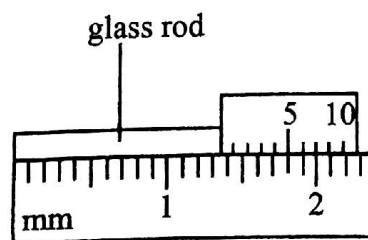


Fig.14.1

- i) Name the instrument used.

..... [1]

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ii) State the length of the glass rod. [1]
.....

iii) State Newton's third law of motion. [1]
.....

iv) Explain what happens when a person sits on a chair in terms of Newton's third law of motion. [2]
.....
.....

b) A force of 300 N pulls an object of mass 60 kg along a horizontal surface.

i) Calculate the acceleration of the object.

[2]

ii) State whether the acceleration value in (c)(i) is higher or lower than the practical value. [1]
.....

- c) Fig.14.2 shows a circuit used to determine the resistance of a wire.

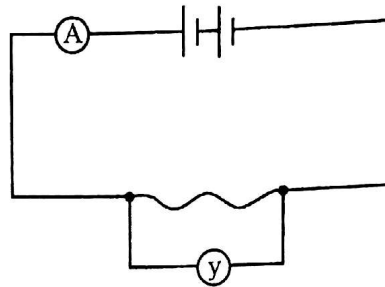


Fig.14.2

- i) Name the instrument y. [1]
-

- ii) State the effect of using a longer wire of the same material and thickness. [1]
-

15. a) i) Name any **three** types of media for signal transmission.

- 1
 2
 3 [3]

- ii) Describe how signals are transmitted in any **one** of the media named in (i). [3]
-

- b) State the function of a decoder.

..... [1]

- c) Describe **three** advantages of e-mail over ordinary mail.

- 1
- 2
- 3

[3]